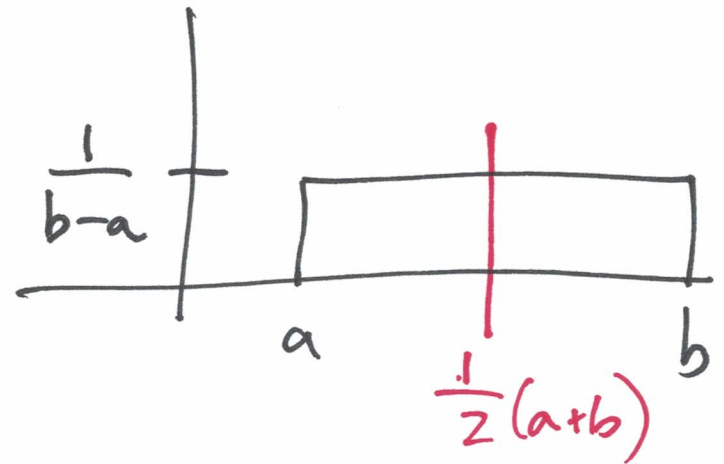


EXPECTED VALUES (CONTINUOUS)

DISCRETE: $E(X) = \sum_{\text{OUTCOMES}} x p_X(x)$

CONTINUOUS: $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$

X UNIFORM on $[a, b]$



$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

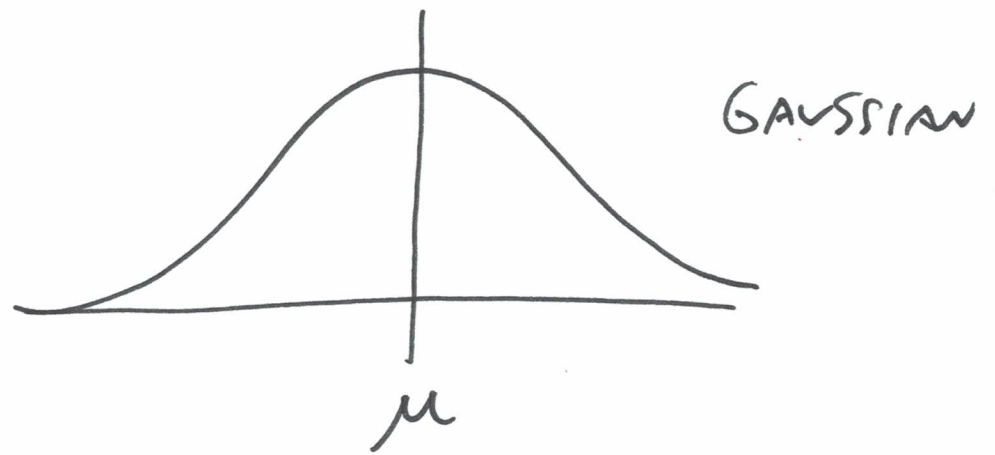
$$= \int_a^b x \frac{1}{b-a} dx$$

$$\frac{1}{b-a} \left[\frac{1}{2} x^2 \right]_{x=a}^{x=b} = \frac{1}{2} \frac{b^2 - a^2}{b-a} = \frac{1}{2} (a+b)$$

SPECIAL CASE OF A SYMMETRIC PDF.

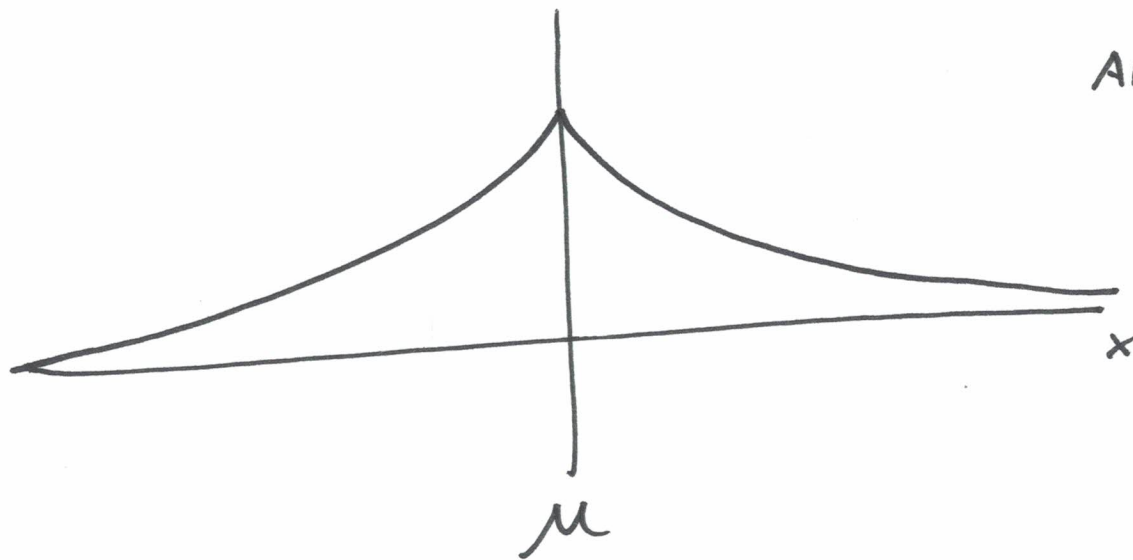
IN THE SAME WAY,

$$E(x) = \mu$$



$$E(x) = \mu_x = \mu$$

WHY TRUE IN GENERAL?



ANY PDF

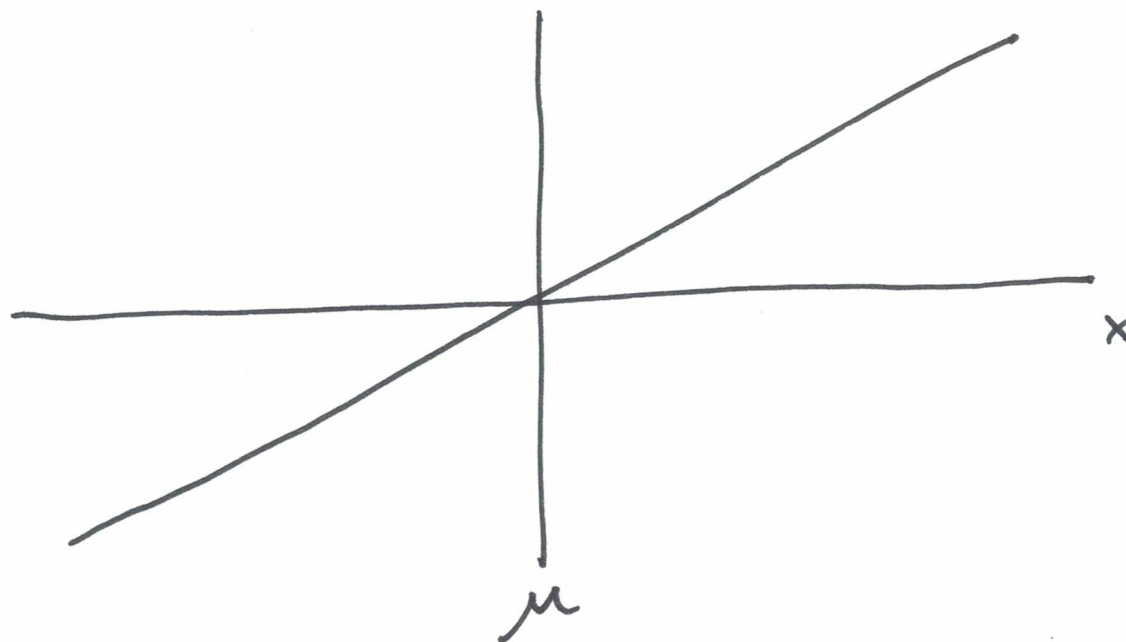
$$f_X(x)$$

SYMMETRIC

AROUND

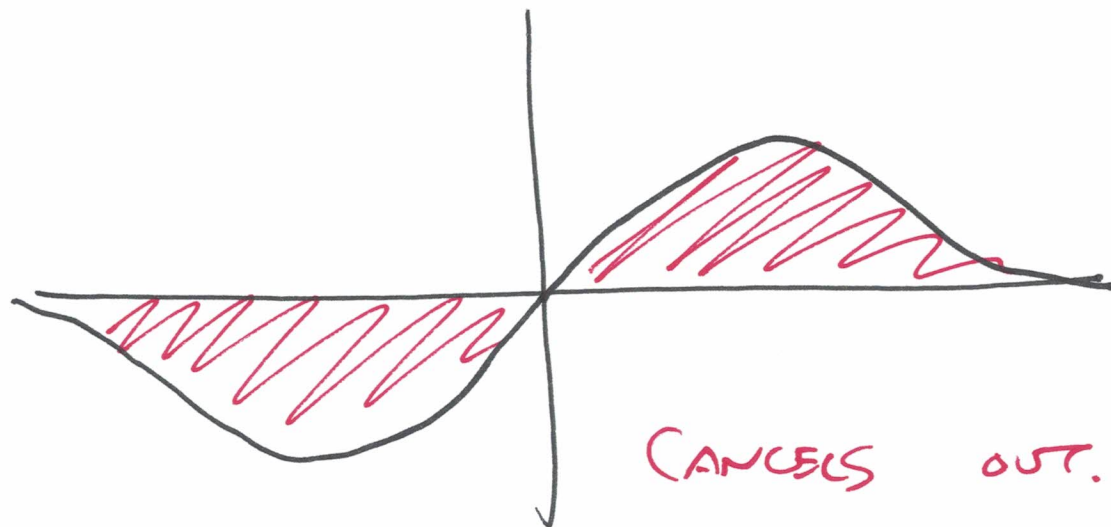
$$x = \mu$$

EVEN



$$x - \mu$$

ODD



$$(x - \mu)f_X(x)$$

ODD.

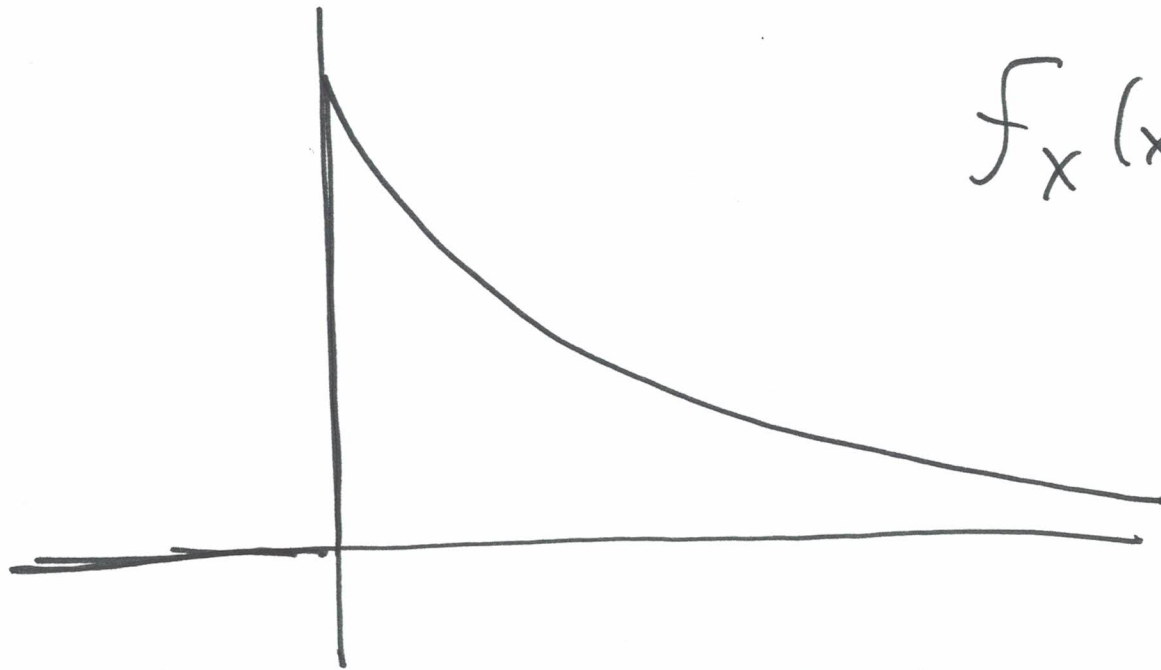
$$\int_{-\infty}^{\infty} (x - \mu)f_X(x) = 0.$$

$$\int_{-\infty}^{\infty} x f_X(x) - \mu \int_{-\infty}^{\infty} f_X(x) = 0$$

$$E(x) - \mu \cdot 1 = 0$$

$$E(x) = \mu.$$

EXPONENTIAL PDF



$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{OTHERWISE} \end{cases}$$

$$E(x) = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} x \cdot \lambda \cdot e^{-\lambda x} dx$$

INTEGRATE BY PARTS... BOOOOO.

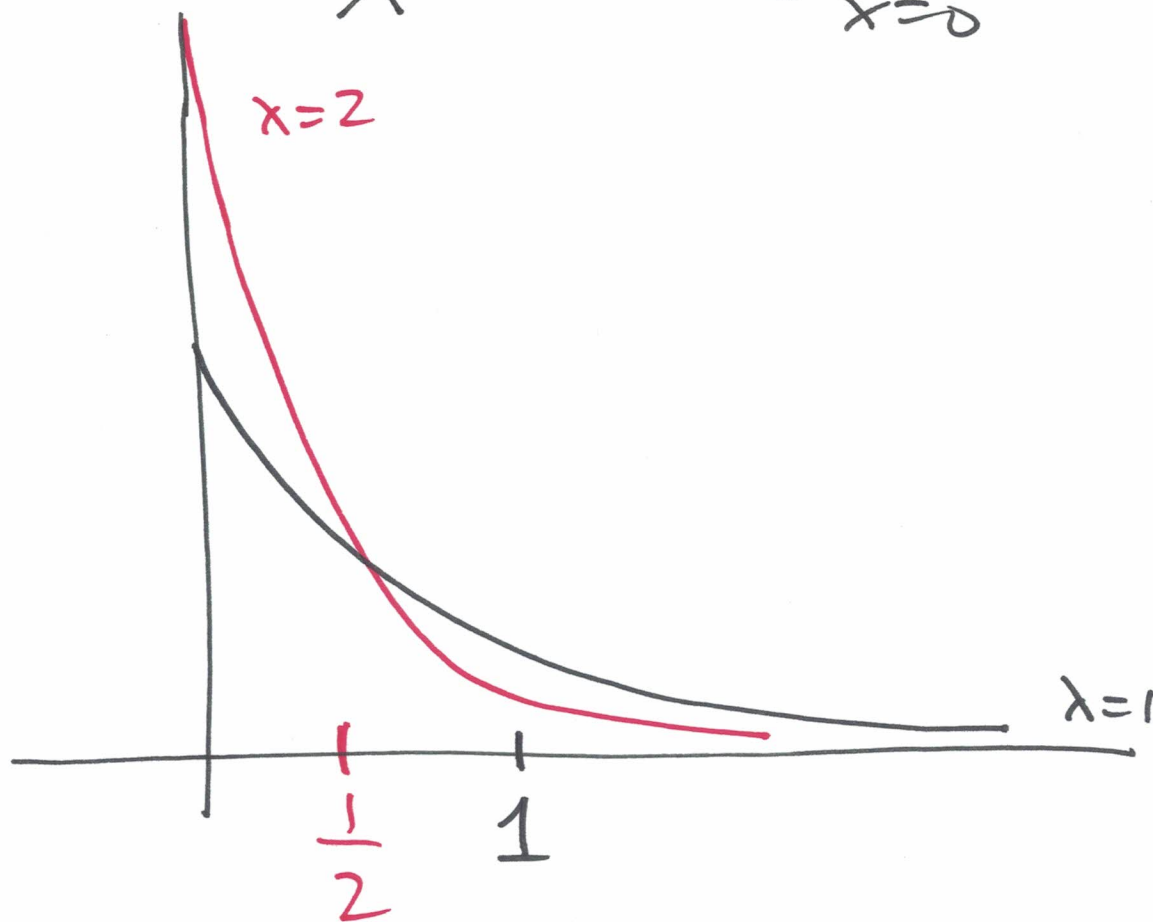
$$= -x e^{-\lambda x} \Big|_{x=0}^{x=\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$\begin{array}{ll} u = x & du = dx \\ v = e^{-\lambda x} & dv = -\lambda e^{-\lambda x} dx \end{array}$$

$$-xe^{-\lambda x} \Big|_{x=0}^{x=\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$-\frac{1}{\lambda} e^{-\lambda x} \Big|_{x=0}^{x=\infty} =$$

$$\boxed{\frac{1}{\lambda}}$$

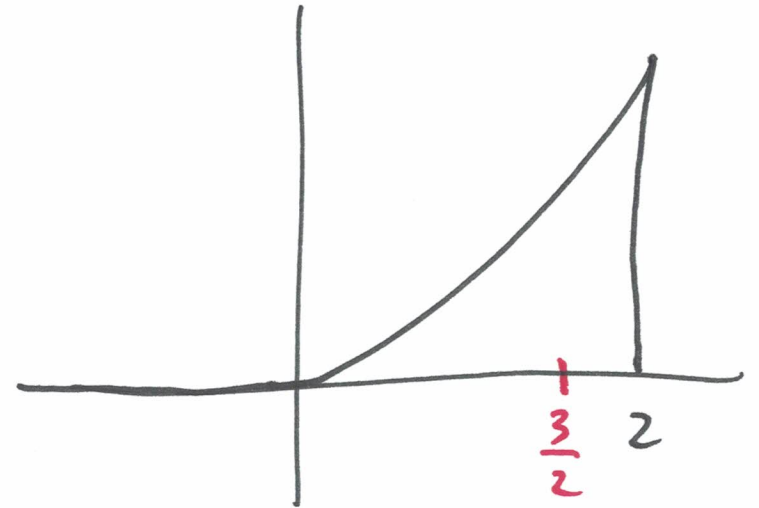


$$\underline{\text{EX}} \quad f_X(x) = \begin{cases} \frac{3}{8} x^2 & x \in [0, 2] \\ 0 & \text{ELSE} \end{cases}$$

$$E(X)?$$

$$= \int_0^2 x \cdot \frac{3}{8} x^2 dx$$

$$= \left. \frac{3}{32} x^4 \right]_{x=0}^{x=2} = \boxed{\frac{3}{2}}$$



OFTEN WE NEED

$$E(g(x))$$

↑
RANDOM.

$$= \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Common:

$$\begin{aligned} \text{Var}(X) &= E((X - \mu_X)^2) \\ &= E(X^2) - (E(X))^2 \\ &= E(X^2) - \mu_X^2 \end{aligned}$$

IN EXAMPLE: $\text{Var}(X)$?

$$E(X^2) = \int_0^2 x^2 \cdot \frac{3}{8} x^2 dx$$

$$= \left. \frac{3}{40} x^5 \right]_{x=0}^{x=2} = \frac{32 \cdot 3}{40} = \frac{12}{5}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= \frac{12}{5} - \left(\frac{3}{2}\right)^2 = \frac{12}{5} - \frac{9}{4} = \boxed{\frac{3}{20}}$$

SANITY CHECK: VARIANCE MUST BE > 0 !

WHAT ABOUT

ANY FUNCTION OF X .
 $E\left(\frac{1}{X^2}\right)?$

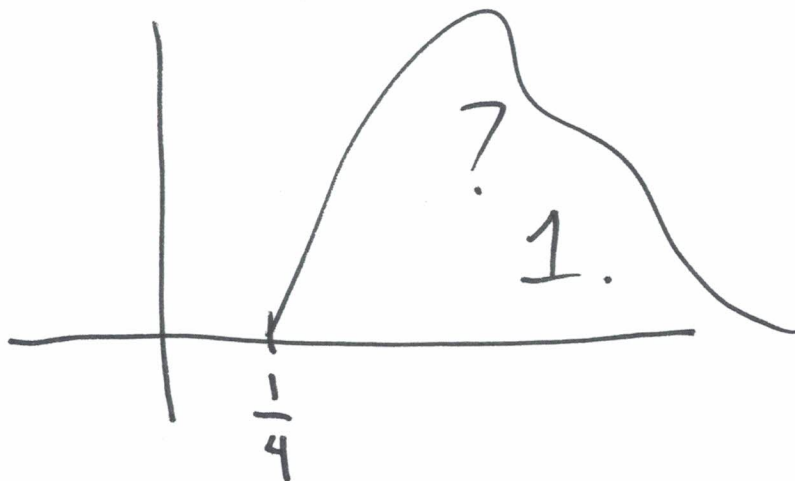
$$= \int_0^2 \boxed{\frac{1}{X^2}} \cdot \boxed{\frac{3}{8} X^2} dx = \int_0^2 \frac{3}{8} dx = \boxed{\frac{3}{4}}$$

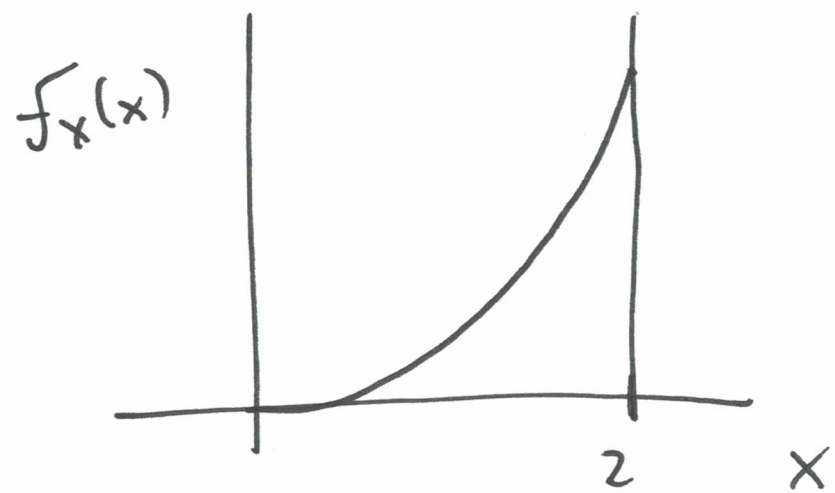
FUNCTION PDF. NUMBER.

PREVIEW: $Y = \frac{1}{X^2}$ A NEW R.V.

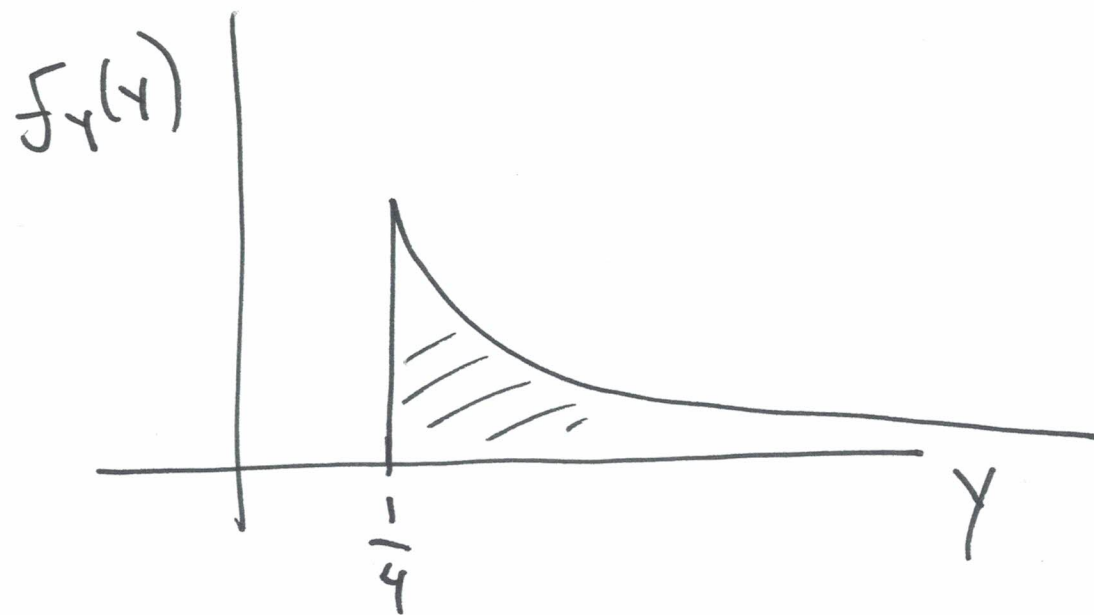
WHAT IS ITS PDF?

$E(Y)$





$$Y = \frac{1}{X^2}$$



PROPERTIES MAKE COMPUTATIONS EASIER:

SAME AS WE HAD BEFORE:

$$E(aX) = aE(X)$$

$$E(X+b) = E(X) + b$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Var}(X+b) = \cancel{\text{Var}(X+b)} \text{Var}(X)$$

Ex X IS GAUSSIAN WITH ~~ME~~
MEAN μ AND STANDARD DEVIATION σ .

COMPUTE $E(2X^2 - 3X + 5)$

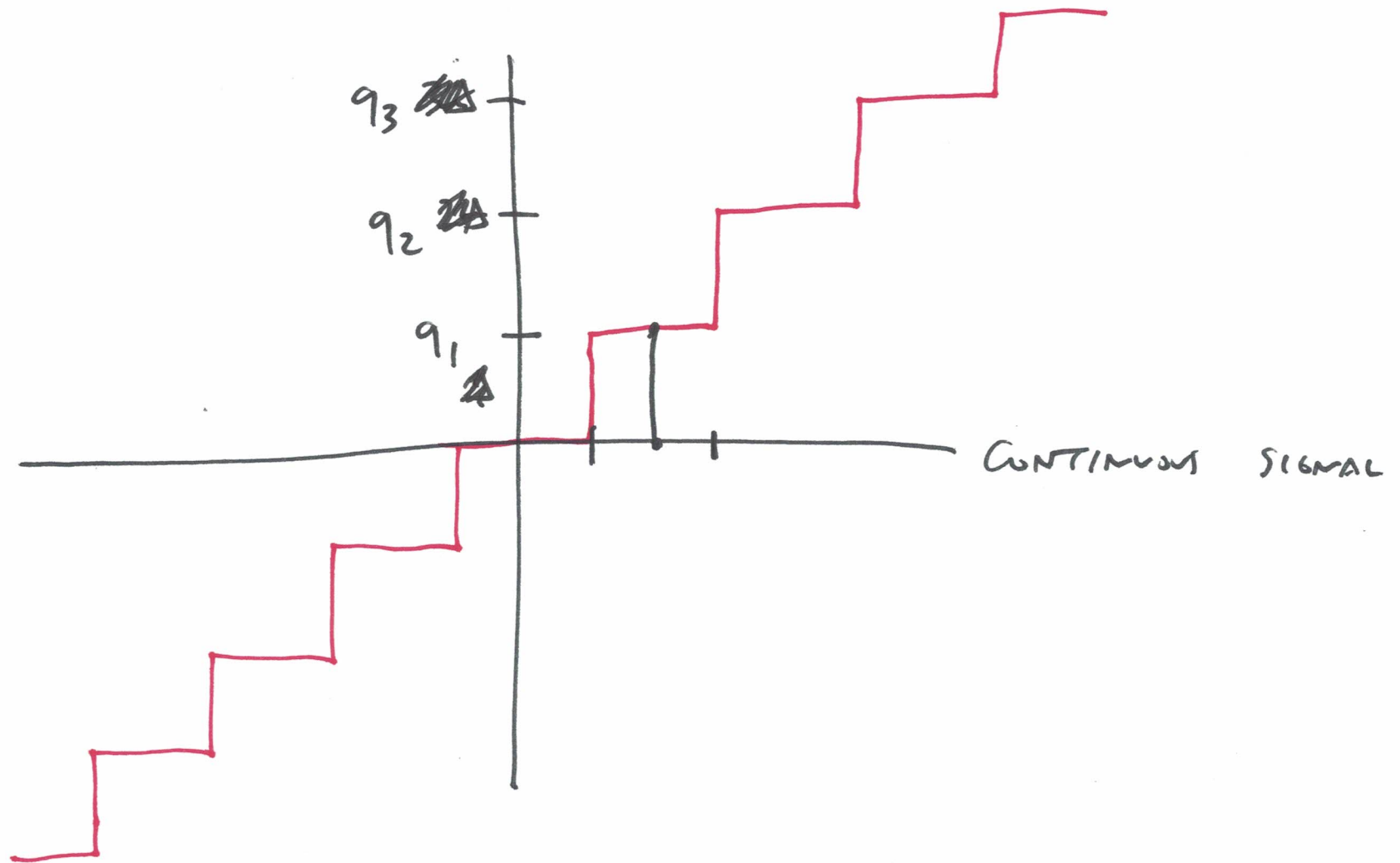
$$= 2E(X^2) - 3E(X) + 5$$

$$2(29) - 3(5) + 5 = \boxed{48}.$$

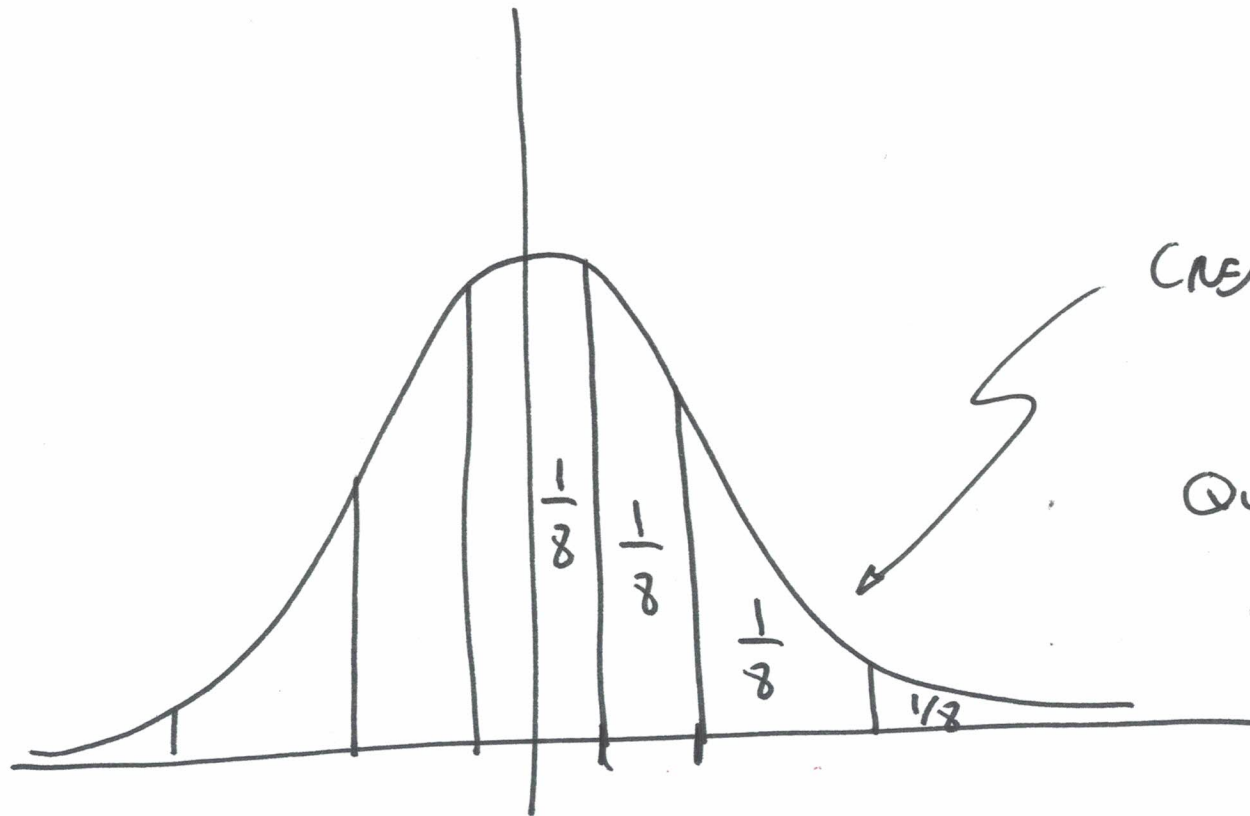
$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = 29$$

SEE PROBABILITY BITES VIDEO FOR
DETAILS ON QUANTIZER DESIGN.



OFTEN, SIGNAL PDF IS NOT UNIFORM.



CREATE EQU-
PROBABLE BINS

QUANTIZER VALUE
IS $E(X | \text{BIN})$